

Short Papers

EasiTia: A Pervasive Traffic Information Acquisition System Based on Wireless Sensor Networks

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Abstract—Traffic information acquisition is often implemented by video cameras or inductive loops, which is expensive or inconvenient from installation and maintenance perspectives. We designed and implemented a pervasive traffic information acquisition system based on wireless sensor networks called EasiTia. Unlike existing solutions, the implementation of the system does not require extra devices in the road infrastructure or vehicle, nor the excavation of the road surfaces. EasiTia can easily be deployed at roadsides. It is of low cost and resource efficient. Our contributions are given as follows: 1) To deal with low signal-to-noise ratios (SNRs) and stochastic disturbances in traffic information acquisition, we proposed and implemented a cross-correlation-based vehicle-detection algorithm. 2) To resolve the problems of data association, vehicle velocity calculation, and vehicle identification, we proposed a collaborative traffic information processing mechanism in the EasiTia system. Based on real road environment experimental analysis, we demonstrate that EasiTia is an applicable and cost-effective candidate for a pervasive traffic information acquisition system.

Index Terms—Collaborative traffic information processing, data association, traffic information acquisition, wireless sensor networks (WSNs).

I. INTRODUCTION

As the foundation of an intelligent transportation system (ITS), traffic information plays an important role in an ITS system. At present, there are many traffic information acquisition systems, such as the inductive loop monitoring system [1], [2], microwave [3], laser [4], infrared [5], and video [6], [7] monitoring system, which have been widely used to obtain traffic information on expressways and main roads in a metropolis. These developed systems can acquire traffic information of high quality in real time. However, nearly all the current traffic information acquisition systems are wired. The detectors are first laid on or embedded in the road; after that, they collect and send data to the traffic management center through wires. Consequently, the deployment and maintenance of these systems is costly, and system scalability is limited. These disadvantages prevent the implementation of comprehensive traffic information support for the ITS. Thus, a pervasive traffic information acquisition system that is of low cost, convenient to deploy, and accurate in performance will be of great significance to ITS implementation in developing countries.

For reasons of low cost, high scalability, and deployment convenience, wireless sensor network (WSN)-based solutions have been re-

search hot spots in traffic acquisition systems [8], [9]. A well-designed WSN-based solution for traffic information acquisition systems should resolve two problems.

- 1) A way to develop a real-time, accurate, and lightweight vehicle-detection algorithm. Vehicle detection is the fundamental system function, because only when vehicles are detected timely and accurately can vehicle information be obtained and more comprehensive traffic information, such as traffic volume, lane occupancy, and vehicle velocity be determined. Most existing [10], [11] methods process raw data with a filter and then use a threshold to detect a vehicle. These methods perform well when the noises and disturbances are relatively low. However, if the signal-to-noise ratio (SNR) is low, these methods will lead to serious degradation in detection quality, such as increases in false alarm and missed detection rates, leading to poor system performance. On the other hand, some constant false alarm rate (CFAR) [12] related methods perform well, but they demand much more computation resource. As a result, the first challenge that should be taken into consideration is to develop an effective detection algorithm with limited resources.
- 2) A way to realize collaborative information processing (CIP). When WSN is applied in the field of traffic information acquisition, because of the unpredictable on-road environment and poor single-node computation capacity, the road-monitoring task should be accomplished in a collaborative way. Furthermore, more comprehensive traffic information cannot be determined from single-node data. To obtain the required information, CIP between multinodes is required. Due to the resource limitations, the mechanism for internode information sharing, processing, and fusion should be explored.

In this paper, we propose a pervasive traffic information acquisition system based on a WSN platform called EasiTia. Our main contributions are given as follows: First, we developed a WSN-based pervasive traffic information acquisition system, which can easily be deployed to various road conditions for traffic information acquisition. Second, to resolve the problems of low detection rate and high false alarm rate in the on-road environment, we proposed a vehicle-detection algorithm based on cross correlation. Third, we developed a collaborative mechanism that calculates vehicle information. With this mechanism, data from multinodes can be fused, and more vehicle information can be obtained.

II. OVERVIEW OF EasiTia SYSTEM

As shown in Fig. 1, EasiTia is composed of several information collection units, and each unit includes several sensor nodes and an information collection node. The sensor nodes are deployed alongside the roads and are set to a power-saving status. When a vehicle is detected, the sensor nodes will be woken up and acquire vehicle information. The collected data will be sent to the corresponding information collection node where the information will be fused and processed. By information fusion, the information collection node will produce comprehensive traffic information for that unit and send it to a base station that is one or more hops away. In the base station, information from the different units will be merged, and useful information can be supplied to potential clients via the network.

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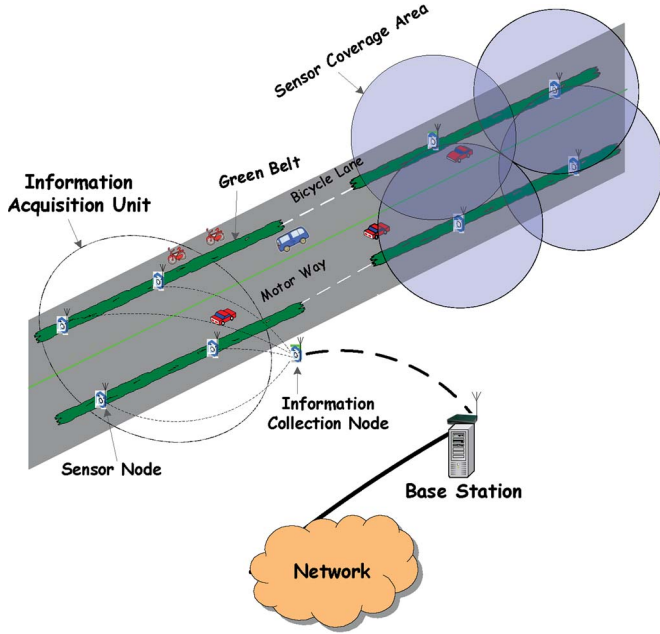


Fig. 1. EasiTia system.

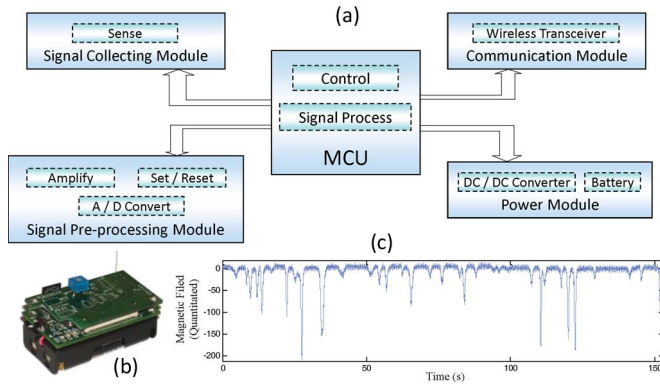


Fig. 2. Easi210 hardware design. (a) Hardware design. (b) Sensor node. (c) Signal sampled by node.

A. Hardware Design

The sensor node is the basic cell of the EasiTia system. In our EasiTia System, we designed a wireless sensor node named EZ210 [13]. As shown in Fig. 2, the sensor node is composed of a signal collection module, a preprocessing module, a microcontroller unit (MCU), a communication module, and a power control module. The signal collection module senses a vehicle and produces an electrical signal. The preprocessing module amplifies the original electrical signal and performs analog-to-digital conversion. It also supplies a reset signal for the sensor. The MCU processes the signal and manages the other modules. The communication module works as a radio transmitter and receiver. The power control module supplies power for other modules.

At present, the sensors used in a traffic information acquisition mainly include the following types: image (camera) sensor, inductive loop detector, acoustic sensor, seismic sensor, magnetic sensor, etc. The image sensor acquires an abundance of information, but it is vulnerable to bad weather and nighttime operation. Furthermore, the image sensor requires more system resources. The seismic sensor and acoustic sensor are vulnerable to noise in deployment environments. Since a vehicle is ferrous, the strength of the earth's magnetic flux around the vehicle will be distorted. Consequently, we use the magnetic sensor HMC1021Z [14] in vehicle detection.

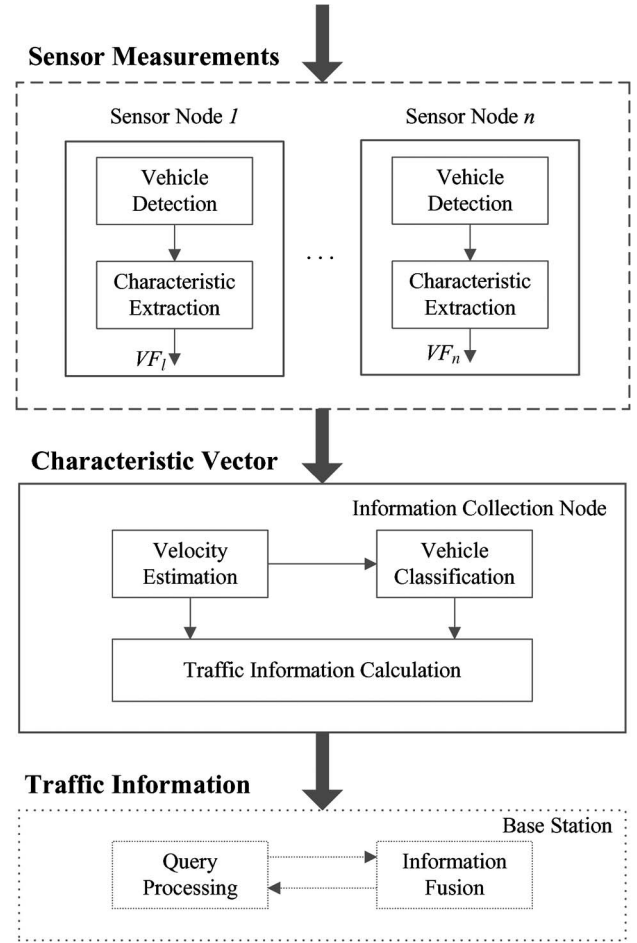


Fig. 3. EasiTia system architecture.

B. Software Design

Traffic information processing algorithm design is the key to the EasiTia system. The information processing consists of sensing, vehicle detection, and vehicle information calculation. Generally, information processing can be designed in a centralized or a distributed way.

However, in a traffic information acquisition system, compared with a distributed processing mechanism, the centralized processing mechanism has the following three drawbacks.

- 1) The processing center may become the bottleneck, because all the original data collected by the nodes are gathered in the center, and the center will undertake too many processing tasks.
- 2) The large amount of data transmission occupies too much bandwidth.
- 3) Sending back original data will consume too much energy, as the communication module is the most energy-wasting module in the node.

For the reasons previously given, we use a distributed processing mechanism in the EasiTia system. Fig. 3 shows the information processing procedure.

Magnetic data are collected by the sensor node. Then, it is processed by the vehicle-detection algorithm to detect the presence of a vehicle. When a vehicle is detected, the sensor node i will calculate the vehicle characteristic vector (feature information) VF_i and send it to the information collection node.

The information collection node listens and processes messages inside a monitoring unit. When VF_i is received, it will first be associated to messages from other nodes, and then, a determination will be

made on whether these messages are generated from the same vehicle object. This process is called data association. After data association, information from different nodes but caused by the same vehicle will be brought together, and a record will be produced, which corresponds to the vehicle information in this unit. Based on these records, more comprehensive traffic information such as vehicle identification and velocity can be calculated.

When the traffic information is produced inside a collection unit, it will be sent to a base station one or more hops away. In the base station, records from different data collection units can be processed by a high-level fusion algorithm. Query and storage services can also be supplied by the base station.

Considering the complicated on-road environment and the limitation of the node resource, effective traffic information acquisition should provide accurate timely vehicle detection and collaborative traffic information processing among the sensor nodes. We will describe the corresponding algorithms in the next section.

III. ALGORITHM DESIGN

A. Cross-Correlation-Based Vehicle-Detection Algorithm

The aim of vehicle detection is to detect a passing vehicle in a fixed region while the detection unit is subject to stochastic noises and disturbances. In this paper, we propose a cross-correlation-based vehicle-detection method called the CD algorithm. The aim of the CD algorithm is to improve the SNR by cross correlation and then to enable an improvement in detection performance.

1) *Algorithm Design:* Without loss of generality, sensor measurement can be expressed as

$$z(t) = s(t) + n(t). \quad (1)$$

In (1), $z(t)$ is a node measurement, and it is composed of two parts: $s(t)$ and $n(t)$. $s(t)$ is the input signal, and $n(t)$ is the measurement noise. Suppose that $h(t)$ is a referential signal; then, the cross correlation of $h(t)$ and $z(t)$ is

$$\begin{aligned} R_{zh}(t) &= \int_{-\infty}^{+\infty} h(\tau) z(t - \tau) d\tau \\ &= \int_{-\infty}^{+\infty} h(\tau) (s(t - \tau) + n(t - \tau)) d\tau. \end{aligned} \quad (2)$$

It is obvious that, if $z(t)$ is the input, $R_{zh}(t)$ is the output. When $h(\tau) = s(t_0 - \tau)$, the maximum spectral energy of $s(t)$ will be retained, so the largest SNR will appear in t_0 . Obviously, a larger SNR will be helpful in detecting useful signal from background noise.

Based on the preceding analysis, the CD algorithm can be described as follows: First, the cross correlation of $z(t)$ and $h(t)$ will be calculated; then, the result $R_{zh}(t)$ will be used to determine the signal type and vehicle-detection output. As we use cross correlation to increase SNR, we can obtain high detection performance with a simple threshold method.

2) *Choice of Referential Signal:* In the CD algorithm, the key problem is to choose a proper referential signal.

Ideally, when a vehicle passes a sensor, the sensor response will reflect the strength change in the magnetic flux. Therefore, we can choose the change curve of magnetic flux as the referential signal.

In the scenario shown in Fig. 4, a sensor is fixed at point C; the vehicle starts from point A, runs at velocity v , and reaches point B

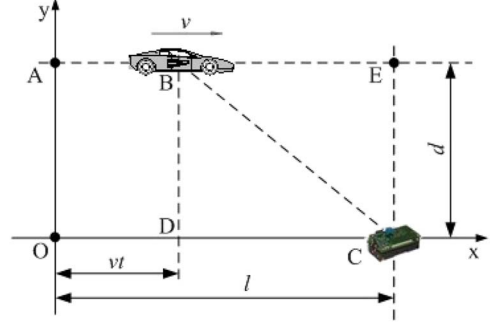


Fig. 4. Sensor and vehicle at time instant t .

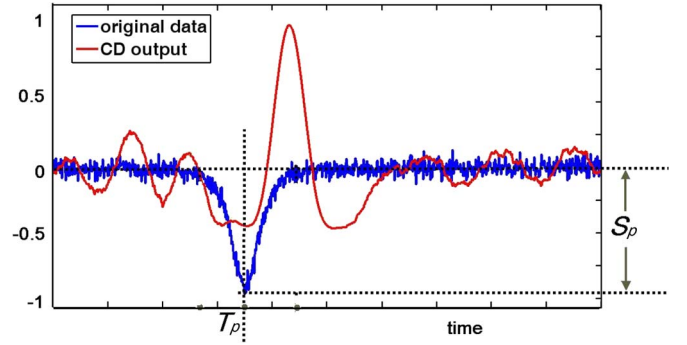


Fig. 5. CD algorithm response.

at time t . Thus, the strength of the earth's magnetic field around the vehicle can be expressed by the following equation [15]:

$$H = m/ur^3 + n \quad (3)$$

where m is the magnetization, r is the distance between the sensor and the vehicle, u is the media constant, and n is the measurement noise.

Then, the distance between B and C is given as follows:

$$l_{BC}(t) = \sqrt{d^2 + (l - vt)^2}. \quad (4)$$

According to (3) and (4), the magnetic strength $H(t)$ is given as follows:

$$H(t) = m / \left(u \cdot (d^2 + (l - vt)^2)^{3/2} \right) + n. \quad (5)$$

Then, the referential signal can be expressed as

$$h(t) = m / \left(u \cdot (d^2 + (l - vt)^2)^{3/2} \right), \quad \sqrt{d^2 + (l - vt)^2} \leq R_s \quad (6)$$

where R_s indicates the maximum sensing range of the sensor. Apparently, an extreme value of $h(t)$ will be obtained if $t = l/v$. According to (6), we have $(l - \sqrt{(R_s^2 - d^2)})/v \leq t \leq (l + \sqrt{(R_s^2 - d^2)})/v$, so the time duration of referential signal 2τ is $2\sqrt{(R_s^2 - d^2)}/v$, where 2τ indicates the time duration for a vehicle passes the detection unit. Consequently, the value of τ conforms to

$$\tau = \sqrt{(R_s^2 - d^2)}/v_{\text{mean}} \quad (7)$$

where v_{mean} indicates the average speed of the vehicle.

3) *Vehicle Feature Calculation:* Once a vehicle is detected by the CD algorithm, vehicle feature information will be calculated by the sensor node. A typical vehicle response is shown in Fig. 5. In this figure, T_p identifies the time instant when the distance between the vehicle and sensor is minimal (peak time instant), and S_p is the response height (peak height) at the time instant T_p .

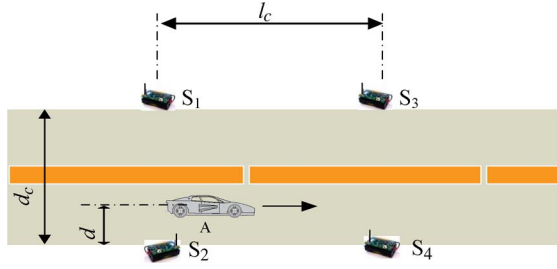


Fig. 6. Simulation scenario.

B. Collaborative Traffic Information Processing

Obviously, it is difficult to gain comprehensive traffic information based only on information provided by a single node. Furthermore, we cannot acquire the comprehensive traffic information simply by accumulating information from several sensor nodes. Consequently, internode CIP needs to be considered. In this section, we utilize CIP to solve three important problems in traffic information acquisition, i.e., data association, velocity calculation, and vehicle identification.

1) *Basic Assumptions and Definitions:* EasiTia is composed of information collection units. In each information collection unit, four sensor nodes S_1 , S_2 , S_3 , and S_4 will be put on the roadside in a rectangular manner, as shown in Fig. 6. Moreover, we assume that the node location is given. Correspondingly, if S_i detects a vehicle presence, it will send vehicle information to an information collection node. Without loss of generality, we assume that the two nodes on the two sides (S_1 and S_2 , or S_3 and S_4 in Fig. 6) will detect a vehicle at the same instant in time. In our on-road experiments, we found that, in practice, the difference in the time instants of detection of these two nodes was very small, and the algorithm performance was only slightly influenced.

Suppose that the distance between nodes on the same side of the road is l_c ; we define the time window T_w as follows:

$$T_w = l_c / v_{\min} \quad (8)$$

where $v_{\min}$ is the minimal velocity, which can be decided by prior knowledge. Obviously, T_w indicates the longest time duration for a vehicle passing the detection unit.

Since we choose the magnetic sensor and the maximum sensing range of the HMC1021Z, R_s is 7 m, we assume that our system is deployed on a bidirectional two-lane road. By adopting multi-detection units, our solution can easily be expanded to a more general road environment.

2) *Data Association:* When a bidirectional two-lane road is assumed, there are three typical scenarios. Scenario 1: for each time frame, only one vehicle appears in the detection unit. Scenario 2: two vehicles appear in the detection unit but run in the same direction. Scenario 3: two vehicles appear in the detection unit but run in opposite directions.

All three scenarios are shown in Fig. 7. In scenario 1, in T_w for any node S_i , only one vehicle is detected. In scenarios 2 and 3, in T_w for node S_i , at least two vehicles will be detected. In all these scenarios, the sensor node should correlate its vehicle measurement in T_w with measurements from the other nodes, which means the data association must be resolved.

In scenarios 2 and 3, it is clear that problems of data association cannot be solved based on measurement from a single node. Therefore, a multinode collaboration-based data association method is proposed to associate measurements with the appropriate vehicles.

In the scenario of two lanes and two directions, data association can be divided into the following two steps: 1) Determine the vehicle lane of this measurement. 2) Determine which vehicle in that lane

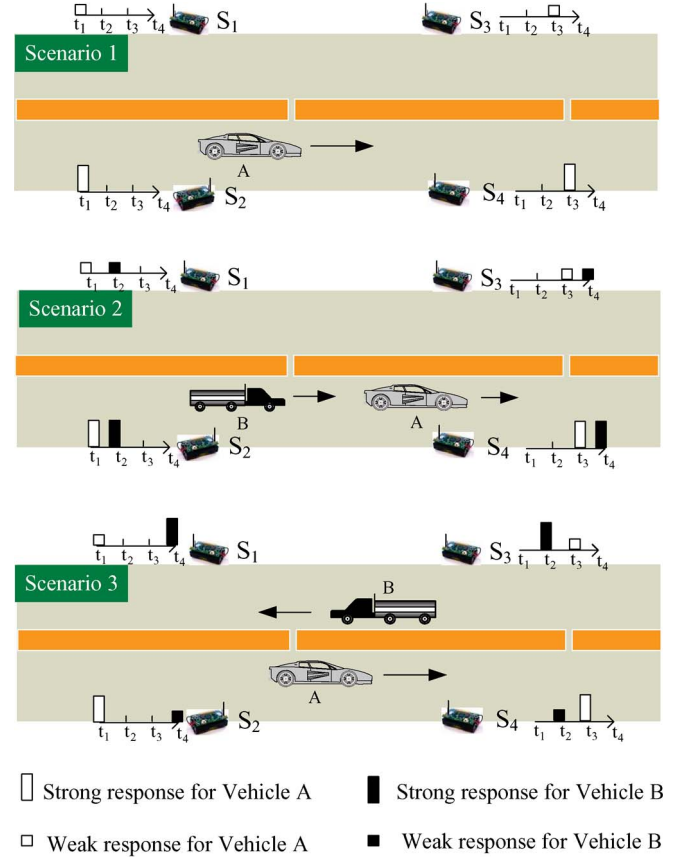


Fig. 7. Typical traffic scenarios. strong response in close side and weak in distant side (Scenario 1: Single vehicle. Scenario 2: Two vehicles, same direction. Scenario 3: Two vehicles, reverse directions).

causes this measurement. Based on this analysis, we use strength and time information to accomplish data association. We determine the vehicle lane information by the measurement strength of the nodes and then accomplish data association through the node's time measurement information.

According to (3), the response strength is in the inverse ratio to r^3 , so the roadside node will produce a strong response when a vehicle appears in the lane closer to it. On the other hand, a weak response will be produced if a vehicle appears in the far lane. This notable property will be used in this paper.

In our system, the distance between in-unit nodes is not very large; correspondingly, the situation in which two vehicles enter and leave the same detection unit can be considered to follow the "First-In First-Out (FIFO)" rule. In other words, vehicle overtaking inside the detection unit is not supposed to occur. Consequently, the time sequence of vehicles passing by can be used to address the data-association issue.

Based on the preceding analysis, we propose a data association algorithm as follows:

When vehicle A passes a detection unit at time instant t , the nodes on both sides i and j will detect this vehicle at the same time. Their response strength measurements $^i Sp_t^A$ and $^j Sp_t^A$ will be produced. Then, a feature vector at the time instant t , i.e., $CH_t^{ij} : (^i Sp_t^A, ^j Sp_t^A)$, can be calculated. In a time window T_w , more than one feature vector may be generated by the node pair i and j . We identify these feature vectors as $CH^{ij} = \{CH_1^{ij}, CH_2^{ij}, \dots, CH_m^{ij}\}$. Meanwhile, another node pair k and l may produce more than one feature vector in T_w . We identify them as $CH^{kl} = \{CH_1^{kl}, CH_2^{kl}, \dots, CH_n^{kl}\}$.

According to the preceding analysis, when CH_t^{ij} is given, based on the strong-weak relationship in the response strength of the feature vector, an alternative subset of CH_t^{kl} , which consists of all the vectors

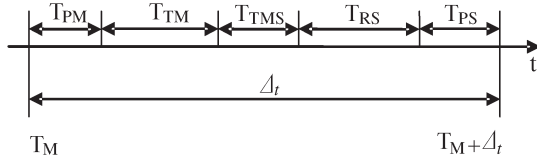


Fig. 8. Time delay in message synchronization.

caused by vehicles from the same lane, can be produced. Meanwhile, if two vehicles run in the same lane, they are traveling in the same direction. Combined with the FIFO rule we assumed, the corresponding vector CH_t^{kl} , which is caused by the same vehicle and results in CH_t^{ij} , can be determined.

3) *Velocity Calculation*: Theoretically, vehicle velocity v can be calculated according to the time duration between the vehicle passing two nodes on the same side. Suppose that, in Fig. 6, the distance between s_1 and s_3 is l_c and the time interval is T_{l_c} ; then, the vehicle velocity can be calculated by the following equation:

$$v = l_c / T_{l_c}. \quad (9)$$

However, to accurately calculate v , the following problems need to be addressed: 1) how to realize accurate synchronization between nodes and 2) the criterion for the distance l_c .

As the nodes do not share the same clock, an accurate synchronization mechanism is required for each detection unit. Here, we introduce a master-slave collaborative synchronization mechanism. Suppose that the master node s_1 is the node that first detects the target and the remaining nodes are the slave nodes.

As shown in Fig. 8, at the beginning of synchronization, the master node broadcasts its local time stamp. The slave node then receives the time stamp T_M and sets its local time to T_S . We have the following definitions as illustrated in Fig. 8: $(T_S - T_M)$ is Δ_t ; T_{PM} is the time interval in which the master node prepares to send the synchronization packet (SynPac for short); T_{TM} is the time interval during which the master node transports the SynPac; T_{TMS} is the interval during which the SynPac is transmitted from the master node to the slave node; T_{RS} is the time interval during which the slave node receives the SynPac; T_{PS} is the time interval in which the slave node processes the SynPac; R_{dT} is the data transporting rate; R_{dR} is the data receiving rate, and N_{Syn} is the length of the SynPac. Thus

$$\Delta_t = T_{PM} + T_{TM} + T_{TMS} + T_{RS} + T_{PS}. \quad (10)$$

Because T_{PM} , T_{TMS} , and T_{PS} are negligible, compared with T_{TM} or T_{RS} , then

$$\Delta_t \approx T_{TM} + T_{RS} \approx R_{dT} \cdot N_{Syn} + R_{dR} \cdot N_{Syn} \approx 2 \cdot R_d \cdot N_{Syn}. \quad (11)$$

The distance between node S_1 and node S_3 in Fig. 7 (expressed as l_c) is predefined before the installation of the monitoring system and is constant since the nodes are static while they are working. On the one hand, the two nodes should be closer to each other, so that no vehicle overtaking occurs in the detection range. On the other hand, as the clocks of the two nodes are not completely consistent, the node distance should be large enough, so that the error caused by the node time difference can be ignored. Thus, the choice of l_c is a tradeoff between the nonovertaking and synchronization differences weakening.

Suppose that our acceptable largest measurement error is ER_v ; then

$$(v_{veh} - v'_{veh}) / v_{veh} \leq ER_v \quad (12)$$

where v'_{veh} is the measurement of vehicle velocity v_{veh} . Δ_t is the error of time synchronization; then, we have

$$v'_{veh} = l_c / ((l_c / v_{veh}) + \Delta_t). \quad (13)$$

According to (12) and (13), if the velocity is maximized ($v_{veh} = v_{max}$), $v_{veh} \cdot \Delta_t \cdot (1 - ER_v) / ER_v$ will be maximized, so l_c should conform to

$$l_c \geq v_{max} \cdot \Delta_t \cdot (1 - ER_v) / ER_v. \quad (14)$$

4) *Target Identification*: In China, it is common for bicycles (non-motor objects) and vehicles (motor objects) to share the same road. Therefore, the ability to identify bicycles is desirable in building an accurate traffic information acquisition system. In general, the ferromagnetic characteristics of a vehicle are much more pronounced than that of a bicycle, and the theoretical strength of response from a vehicle is usually greater than that from a bicycle. This information about the strength of the magnetic response can be used in vehicle identification. However, the strength of magnetic response is also related to the distance between the observation node and the object. The strength of the response will decay with the increment in distance. Therefore, the strength of the response caused by a distant vehicle may approximate that of a closer bicycle. Velocity is another useful source of information in vehicle identification. The velocity of a vehicle is usually greater than the velocity of bicycle, and only in special cases such as a car slowing down to a stop or traffic jam may it be slower than a typical bicycle.

According to the aforementioned analysis, it can be seen that, if only one type of physical information is used, the identification accuracy might not be satisfying. However, if multiple types of information are fused, particularly when the types of information are complementary, the identification accuracy may be improved. In this paper, the Dempster-Shafer Theory of Evidence [16] (DST) is adopted to identify the vehicle, and we take these two types of information as evidences.

The foremost issue in DST reasoning is the construction of a *basic belief mass model*. Based on the statistical analysis of the large quantities of actual on-road measured data, we construct a simple but approximate *basic belief mass model*.

Based on the statistical analysis, the lower limit of the response strength of a vehicle (VS_{low}) and the upper limit of the response strength of a bicycle (BS_{high}) can be obtained. VS_{low} may be caused by a mini car in the far lane, and BS_{high} may be caused by a bicycle in the near lane. BS_{high} is usually higher than VS_{low} . When Sp (response strength) is lower than VS_{low} , the detected object is more likely to be a bicycle. Thus, we assign bicycle a high mass value of 0.9 and vehicle a low mass value of 0.1. When Sp is greater than BS_{high} , the detected object is more likely to be the vehicle. Thus, in this case, we assign bicycle a low mass value of 0.1 and vehicle a high mass value of 0.9. However, when Sp is between VS_{low} and BS_{high} , it might be either a vehicle or a bicycle. Therefore, we simply assign both of them to be 0.5. The mass model for response strength is shown in

$$m_{Sp}(\text{bicycle}) = \begin{cases} 0.9, & Sp < VS_{low} \\ 0.1, & Sp > BS_{high} \\ 0.5, & \text{Others} \end{cases}$$

$$m_{Sp}(\text{vehicle}) = \begin{cases} 0.1, & Sp < VS_{low} \\ 0.9, & Sp > BS_{high} \\ 0.5, & \text{Others} \end{cases} \quad (15)$$

As for velocity, similar rules are built based on the lower limit of the velocity of the vehicle VV_{low} and the upper limit of the velocity of the bicycle BV_{high} . The difference is that, when the velocity of detected object is between VV_{low} and BV_{high} , we assume that the probability of it being a bicycle linearly decreases with the increase in velocity while the probability of it being a vehicle linearly increases.

TABLE I
SIMULATION PARAMETERS

Simulation Parameters	Parameter Value
Maximum vehicle speed V_{max}	14m/s
Minimum vehicle speed V_{min}	5m/s
Average vehicle speed v_{mean}	9.5m/s
Vehicle-sensor distance d	3.5m
Variance of measurement noises n	11.5 UG^2 *
Maximum sensing range R_s	7m
Road width d_c	6m
Sampling frequency f_s	100Hz
Synchronization time difference Δ_t	60 ms
Largest velocity measurement error ER_v	15%
Minimum response strength of vehicle VS_{low}	11.83 UG *
Maximal response strength of bicycle BS_{high}	38.56 UG *
Minimum velocity of vehicle VV_{low}	5 m/s
Maximum velocity of bicycle BV_{high}	8.33 m/s

* UG is the HMC1021 resolution, $1UG=85\mu Gauss$

The mass model for velocity is shown in

$$m_v(\text{bicycle}) = \begin{cases} 0.9, & v < VV_{low} \\ 0.1, & v > BV_{high} \\ 0.9 - \frac{0.8 \cdot (v - VV_{low})}{BV_{high} - VV_{low}}, & \text{Others} \end{cases}$$

$$m_v(\text{vehicle}) = \begin{cases} 0.1, & v < VV_{low} \\ 0.9, & v > BV_{high} \\ 0.1 + \frac{0.8 \cdot (v - VV_{low})}{BV_{high} - VV_{low}}, & \text{Others} \end{cases} \quad (16)$$

According to the DST combination rules, we can get combination evidence through the fusion of the response strength evidence and the velocity evidence. After fusion, the masses for the bicycle and the vehicle can be calculated, respectively. If the mass of the bicycle is larger than that of the vehicle, we would identify the detected object as the bicycle; otherwise, it is the vehicle.

IV. SYSTEM EVALUATION

A. Simulation Settings

Our on-road experiments were carried out on a bidirectional two-lane road, as shown in Fig. 6. Four Easi210 nodes equipped with HMC1021Z were put on the roadside. The detection unit reported its measurements when a vehicle passed through the detection unit.

As we have mentioned before, our proposed system is low in cost for pervasive deployment. In this experiment, one detection unit, including four sensor nodes and an information collection node, costs less than \$250.

Guided by statistics on actual on-road measured data, we adopt the following values for the parameters. These parameters will be used in the personal computer simulation and on-road experiments.

According to Table I, some detailed parameters can be calculated. Based on (14), the distance between nodes on the same side of the road should conform to $l_c \geq 4.76$ m; so here, we choose $l_c = 6$ m. Based on (8), the window duration T_w is 1.2 s. According to (7), the time duration of the referential signal can be calculated, i.e., $\tau = \sqrt{(R_s^2 - d^2)}/v_{mean} \approx 0.7$ s. Since we set the sampling rate f_s to be 100 Hz, the length of the referential signal is $2 \times \tau \times f_s = 140$. Then, according to (6), we can calculate the normalized referential signal $h(t) = -42.875 \times (12.25 + x(t)^2)^{-3/2}$, $|x(t)| \leq 7$, where $x(t) = l - vt$ is l_{DC} in Fig. 4.

B. On-Road Experiments

The on-road experiment is shown in Fig. 9. To verify the basis of our experiment, we used a video camera to capture visual images, so that

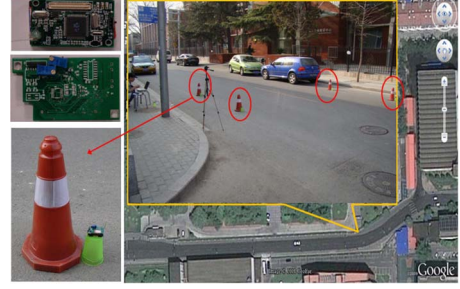


Fig. 9. On-road experiments.

TABLE II
VEHICLE DETECTION RESULT OF ON-ROAD EXPERIMENT

Group	Count by video	Count by CD Algorithm	Detection accuracy (%)
1	10	9	90.00
2	10	10	100
3	16	15	93.75
4	4	4	100
5	5	5	100
6	13	13	100
7	12	11	91.67
8	10	9	90.00
9	11	9	81.82
10	13	12	92.31
Total	104	97	93.27

the detection accuracy could be established by the video. The video captures image at 24 frame/s, and, by counting the frames as a vehicle passes through the detection unit, the real vehicle velocity can be obtained. In this section, we will show the results of the on-road tests on vehicle detection, velocity calculation, and vehicle identification.

1) *Detection Experiments:* Here, we verify the CD algorithm via on-road experiments. The CD algorithm was implemented on Easi210 nodes, and an on-road detection experiment was conducted.

The detection result is shown in Table II. In total, ten groups of detection test were completed. The worst result was over 80%, whereas the best was 100%. The overall detection quality was about 90%.

To compare the performances of the different detection methods, we processed on-road sensor data with several commonly used methods, including the direct threshold measurement method [10] [using thresholds on the raw data (DT)] and the finite impulse response (FIR)-based threshold measurement method [11] [using the thresholds on the data processed by FIR (FIRT)]. The actual number of vehicles is ascertained by watching the synchronized video. In the CD simulation, the data are processed by the CD algorithm with two thresholds, i.e., 0.3(low) and 0.4(high). In the DT simulation, the data are processed by the threshold algorithm. The value of the threshold is 10UG(low) and 20UG(high), using the unit (HMC1021 resolution), below the background noise strength. In the FIRT simulation, the data are first processed by a low-pass filter (the cutoff frequency of the filter is 4 Hz); then, two thresholds, i.e., 10UG(low) and 20UG(high), are used to detect the appearance of the vehicle. According to the results shown in Table III, when the low threshold is adopted, the detection rate of the DT and FIRT is relatively high, but the false alarm rate is also relatively high. When the high threshold is adopted, although the false alarm rate of the DT and FIRT can be reduced, their detection rate also decreases. Referring to our proposed CD algorithm, we see from this table that the CD algorithm is not sensitive to the threshold, and its false alarm rate is very low (0 when the threshold is 0.3 and 0.4). Therefore, we can conclude that the CD algorithm outperforms the commonly used detection methods. Furthermore, the results indicate that the CD algorithm is not threshold sensitive.

TABLE III
ORIGINAL MAGNETIC DATA FOR SIMULATION

Type	Threshold	Detected	Missed	False alarm (FA)	Detection rate (%)	FA rate (%)
CD	low	36	3	0	92.3	0
	high	35	4	0	89.7	0
DT	low	69	2	32	94.8	82.1
	high	39	14	14	64.1	64.1
FIRT	low	43	7	11	82.1	25.6
	high	26	13	0	66.7	0

TABLE IV
VELOCITY CALCULATION RESULT OF ON-ROAD EXPERIMENT

Group	Average Velocity (km/h)	Average Reference Velocity (km/h)	Average Accuracy of Velocity
1	40	39	87%
2	31	31	94%
3	23	24	93%
4	12	11	88%

TABLE V
VEHICLE IDENTIFICATION RESULTS OF DIFFERENT APPROACHES

	Bicycles identified	Vehicles identified	Bicycle identification accuracy	Vehicle identification accuracy	Overall identification accuracy
R_{vi}	26	11	83.87%	26.83%	51.39%
V_{vi}	27	33	87.10%	80.49%	83.33%
D_{vi}	28	36	90.32%	87.80%	88.89%

2) *On-Road Experiments of Velocity Calculation*: We conducted four groups of tests, and each test lasted 10 min.

In all four tests, nine vehicles appeared in group 1, 11 vehicles in group 2, five fast bicycles in group 3, and five slow bicycles in group 4. The results of the average velocity for each group are listed in Table IV. In groups 1 and 2, it can be seen that, when the velocity was relatively high, the estimation error was large. The reason is that, in this case, the error of time duration calculation was relatively large. The average accuracy of the velocity measurement was estimated to be 90%.

3) *Vehicle Identification*: Based on the constructed masses of the response strength and the velocity above, we implemented DST-based vehicle identification (D_{vi} method). To compare the difference between identification results before and after fusion, we also implemented vehicle identification only according to the response strength (R_{vi} method) and only according to the velocity (V_{vi} method). In R_{vi} , the average of the mean vehicle response strength and the mean bicycle response strength was taken as the threshold. We also applied similar operations to V_{vi} .

In our on-road simulation, the sample size of the detected objects was 72, of which 31 were bicycles and the rest objects were vehicles. The identification results are shown in Table V.

According to Table V, if the vehicle identification is accomplished only based on response strength, the accuracy will be low, i.e., about 51.39% in our experiment. The reason is that the magnetic response strength of a vehicle is decided by not only the magnetic feature of the vehicle itself but also the distance between the vehicle and sensors. The uncertainty of response strength leads to the low identification accuracy.

Compared with strength-only vehicle identification, the identification accuracy based only on speed is about 83.33%. The reason is that velocity information is more reliable in identification, and vehicle is commonly much faster than bicycle. A better identification, with identification accuracy up to 88.89%, is achieved by the DST method. In particular, the vehicle-identification accuracy is greatly improved.

The reason is that two kinds of complementary information are taken into consideration.

V. CONCLUSION AND FUTURE WORK

In this paper, we have designed and implemented a pervasive traffic information acquisition system based on WSNs. Unlike traditional video cameras or inductive loops, our solution is of low cost and resource efficient. It can easily be deployed at the roadside. Furthermore, the system implementation does not require the excavation of the road surface and extra devices in the road infrastructure. Compared with radio-frequency identification or in-vehicle traffic information acquisition systems, no device on a vehicle is required by our system. Thus, our proposed system is more adaptable and scalable.

In the future, we will collect enough traffic data, so that we can develop intelligent algorithm to improve the system performance. Meanwhile, we plan to utilize the collected data to enable our proposed approach to deal with more comprehensive traffic scenarios. Furthermore, we plan to use more types of sensors, e.g., acoustic or image sensor, to gain new physical information. By fusion of multimodal sensor information, we will investigate more comprehensive and accurate traffic information systems.

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